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Bill Scanning Based Reward & Intelligence System

Study Project Report

Phase "2"

Presented By **Team ARAJ** – Group (120)
Advised By **Prof. Uma Sankara Rao**

Project Details:

Course Title	B.Sc. Computer Science – BITS Pilani Digital	
Project Title	Bill Scanning Based Reward & Intelligence System	
Phase	2	
Group No.	120	
Gropu Name	Team ARAJ	
Number of members	4	
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Group Advisor	Prof. Uma Sankara Rao	
Submission Date	21 | February | 2026	

Team ARAJ

Who We Are

Team ARAJ, Group 120, is a four-member student team from the B.Sc. Computer Science program (BITS Pilani Digital). This study project is completed as part of our final-year academic requirements, emphasizing real-world problem identification and the creation of feasible, technology-driven solutions.

Our chosen project area is Artificial Intelligence and Machine Learning (AI/ML), focusing on data extraction, pattern recognition, and converting unstructured data into meaningful, privacy-compliant intelligence.

Bill Scanning–Based Reward & Intelligence System

Team ARAJ — **A**shfaaq, **R**anjeet, **A**rpan, **J**yoti

Phase 2: Design & Proof of Concept

1. Introduction

1.1 Purpose of Phase 2

The objective of Phase 2 is to translate the Phase-1 problem definition into a concrete and implementable system design.

This phase focuses on:

- Designing a scalable AI-driven receipt intelligence system
- Defining complete functional and non-functional requirements
- Creating system architecture (frontend, backend, ML, database, DSP layer)
- Demonstrating technical feasibility via a Proof of Concept (PoC)

The system aims to transform raw purchase receipts into structured transaction intelligence and monetize anonymized behavioral cohorts via DSP-compatible exports.

Summary

Phase 2 converts problem definitions into a scalable AI architecture. It focuses on technical requirements, ML-driven receipt processing, and a PoC to transform raw data into monetizable, DSP-compatible behavioral cohorts..

1.2 Scope of Phase “2”

Phase 2 includes:

1. High-level and low-level system architecture
2. ML system design for OCR, fraud detection, segmentation, and intent modeling
3. Reward engine behavioral design
4. Database schema and feature store architecture
5. Data flow and processing pipeline
6. DSP integration design
7. A working Proof of Concept validating receipt ingestion and reward processing

2. System Overview

2.1 Product Perspective

The system is a standalone cloud-native platform consisting of:

→ ● Mobile application (iOS/Android)

→ ● Backend microservices

→ ● ML inference services

→ ● Feature store and analytics layer

→ ● Reward engine

→ ● DSP export interface

It interacts with:

→ ● Users uploading receipts

→ ● External email/SMS APIs

→ ● Cloud storage services

→ ● DSP platforms via hashed cohort exports

Deployment environment:

→ ● Cloud (AWS/GCP)

→ ● Mobile-first architecture

2.2 Major System Functions

The system supports:

1. User registration and authentication
2. Receipt ingestion (image, PDF, email, SMS)
3. OCR-based structured extraction
4. Merchant normalization and SKU classification
5. Fraud detection and trust scoring
6. User behavioral feature aggregation
7. Dynamic reward computation
8. Tier and streak management
9. Cohort building and audience export for DSPs
10. Consent management and privacy compliance

2.3 User Classes and Characteristics

End Users

- Upload receipts
- Earn rewards
- Track tier progression
- Consent to anonymized data usage

Administrators

- Monitor system health
- Manage reward rules
- Oversee fraud detection
- Configure DSP exports

External Systems

- Email APIs
- SMS APIs
- DSP platforms
- Clean room partners

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3. Functional Requirements:

→ **FR1:** The system shall allow users to register and authenticate securely.
→ **FR2:** The system shall allow receipt uploads via image, PDF, email, and SMS.
→ **FR3:** The system shall extract structured transaction data using ML-based OCR.
→ **FR4:** The system shall normalize merchant names and classify SKUs.
→ **FR5:** The system shall compute fraud risk scores before reward issuance.
→ **FR6:** The system shall compute dynamic rewards based on configurable rules.
→ **FR7:** The system shall maintain user tiers and streak progress.
→ **FR8:** The system shall aggregate behavioral features in a feature store.
→ **FR9:** The system shall generate user segments using unsupervised clustering.
→ **FR10:** The system shall compute intent probabilities using supervised models.
→ **FR11:** The system shall build anonymized cohorts for DSP export.
→ **FR12:** The system shall log and enforce user consent policies.

4. Non-Functional Requirements

4.1 Performance Requirements

→ ● Receipt processing latency: < 5 seconds (asynchronous acceptable)
→ ● Reward calculation: < 500ms
→ ● Support for 1M+ users horizontally scalable
→ ● ML inference auto-scaling via Kubernetes

4.2 Security Requirements

→ ● JWT-based authentication
→ ● OAuth2 support
→ ● AES-256 encryption at rest
→ ● TLS encryption in transit
→ ● PII stored in encrypted vault
→ ● Hashed identifiers for DSP export

4.3 Usability Requirements

→ ● Simple receipt upload flow
→ ● Clear reward dashboard
→ ● Tier progress visualization
→ ● Minimal cognitive friction

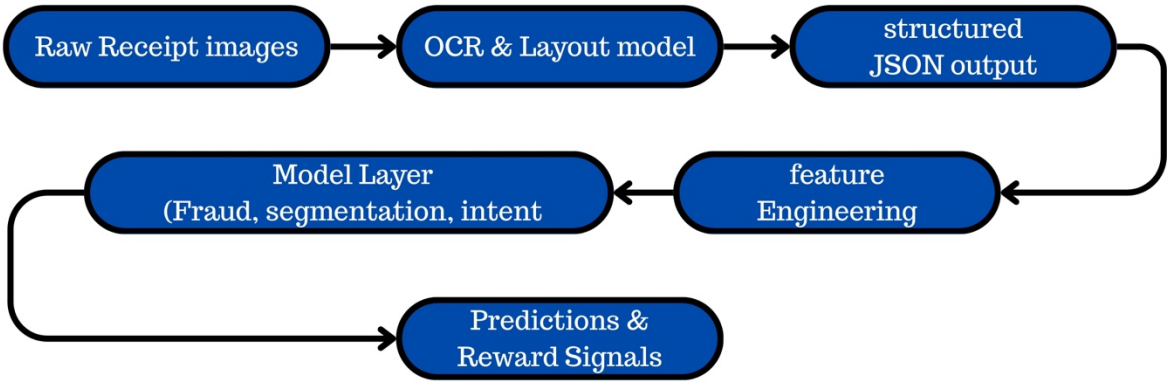
4.4 Scalability and Maintainability

→ ● Microservices-based architecture
→ ● Event-driven processing via Kafka
→ ● Feature store abstraction
→ ● Model versioning via MLflow
→ ● Blue/green deployments

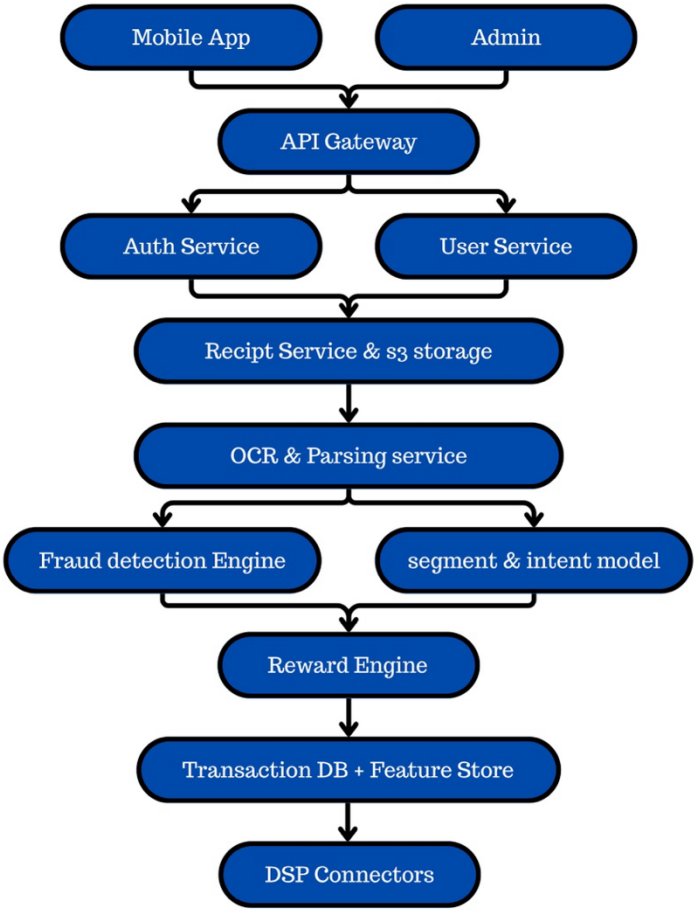
5. System Architecture and Design

5.1 System Architecture Diagram

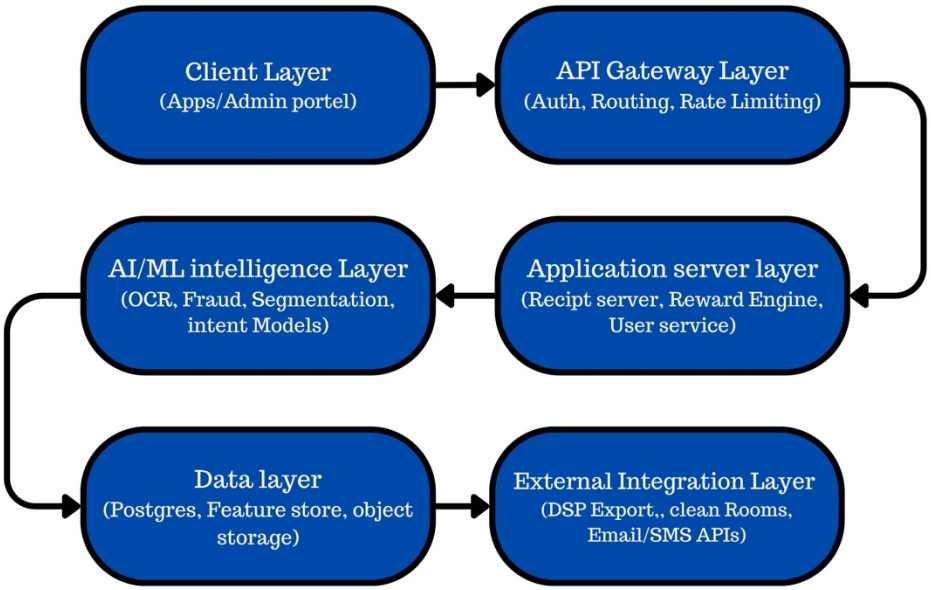
ML Pipeline Architecture:



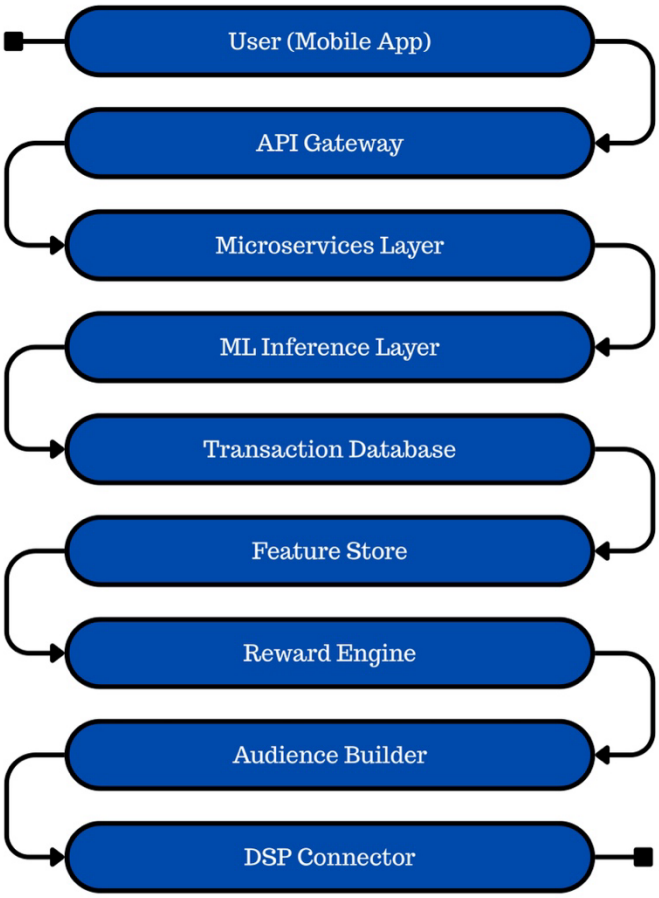
Low Level System Design Architecture:



High Level System Layered Architecture:



High-Level Architecture:



Expalination

The system follows an event-driven microservices architecture. Receipt uploads trigger asynchronous ML processing. Extracted data is structured, stored, enriched, and used for reward calculation and audience generation.

5.2 Module-wise Design

1. Authentication Service

→ OAuth2 support
→ Social login
→ JWT issuance
→ Refresh token rotation
→ Consent tracking

Storage:

- Postgres (user table)
- Redis (session cache)

2. Receipt Ingestion Pipeline

Step 1: Upload

- Image / PDF → S3
- Metadata → Postgres
- Event → Kafka

Step 2: Preprocessing Worker

- Image normalization
- Rotation detection
- Noise removal
- Store processed copy

3. OCR & Extraction Service

ML Inference Layer

Model Stack:

→ Vision Transformer (fine-tuned)
→ LayoutLMv3 (structured extraction)
→ Token classification head

Deployment:

- GPU-enabled inference pods
- Autoscaling via HPA

Output:

Structured JSON:

```
{
 "merchant": "Reliance Fresh",
 "date": "2026-02-18",
 "total": 1250.00,
 "items": [
   {"name": "Aashirvaad Atta", "price": 450},
   {"name": "Milk", "price": 60}
 ]
}
```

4. Merchant Resolver

Normalizes merchant names via embeddings.

Technique:

→ Sentence embeddings (SBERT)
→ Approximate nearest neighbor search
→ Merchant registry DB

Prevents:

→ Duplicate merchants
→ Name variants

5. Fraud Detection Service

Models:

- Isolation Forest
- Velocity-based heuristics
- Graph anomaly detection

Real-time scoring before reward issuance.

6. Feature Store Service

Aggregates user-level behavioral metrics.
Central ML foundation.
Features:

→ Spend recency
→ Frequency
→ Category entropy
→ Premium index
→ Brand loyalty
→ Spend volatility
→ Time-of-day purchase pattern

Tech:

- Feast
- BigQuery/Snowflake
- Redis for online serving

7. Reward Engine:

Computes dynamic incentives based on configurable rules.

Reward Calculation Service

Inputs:

→ ● Transaction value
→ ● Category multiplier
→ ● Tier
→ ● Streak
→ ● Fraud score

Processing:

→ ● Dynamic multiplier calculation
→ ● Randomized reward boost probability
→ ● Tier progression logic

Behavioral Loop Mechanisms

→ Streak counter
→ Tier ladder
→ Milestone bonus
→ Variable reward distribution

State stored in:

→ Redis (real-time)
→ Postgres (persistent)

8. Segmentation Engine

Clusters users via unsupervised ML.

Unsupervised:

→ HDBSCAN
→ K-Means
→ Autoencoder embeddings

Outputs:

Segment IDs per user.

9. Intent Prediction Service

Predicts future purchase probability using XGBoost.

Supervised:

→ XGBoost
→ LightGBM
→ LSTM (future sequence modeling)

Prediction:

```
Probability(user buys category X in 14 days)
```

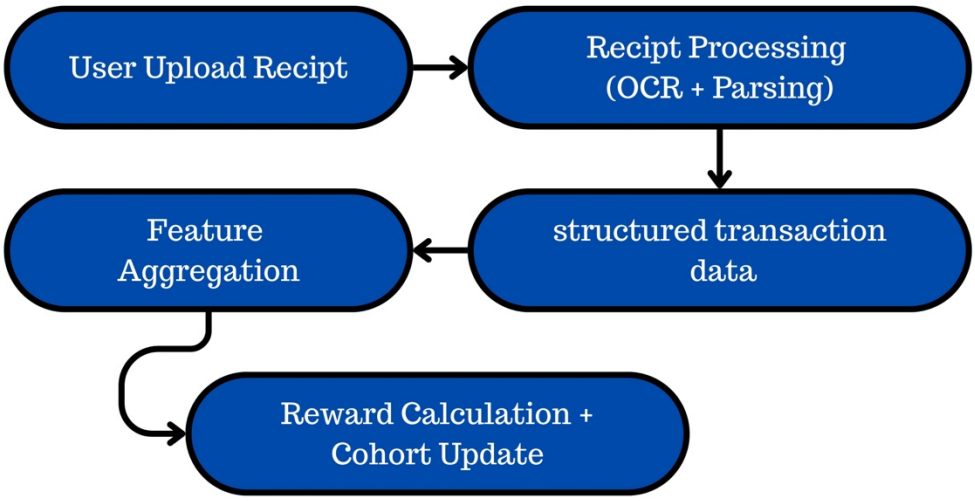
Used for:

→ Reward boosts
→ Coupon recommendations

10. Audience Builder

Constructs DSP-compatible anonymized cohorts.

5.3 Data Flow Design:



1. User uploads receipt
2. Receipt stored in S3
3. Event sent to Kafka
4. OCR service extracts structured data
5. Data stored in Postgres
6. Feature store updated
7. Fraud score computed
8. Reward calculated
9. Segment updated
10. Cohort eligibility evaluated

5.4 Database Design

Key Entities:

→ Users
→ Receipts
→ Receipt_Items
→ Merchants
→ User_Features
→ Fraud_Scores

Rewards:

→ Reward_Transactions

→ User_Segments

→ Audience_Cohorts

→ Consent_Logs

Relationships:

- One User → Many Receipts
- One Receipt → Many Receipt_Items
- One User → One Feature Vector
- One User → One Segment Assignment
- One Cohort → Many Users

6. Technology Stack and Justification

- **Frontend**
  Flutter / React Native
  → Cross-platform efficiency and rapid iteration.

- **Backend**
  Python (FastAPI)
  → High ML compatibility and async performance.

- **Database**
  PostgreSQL
  → Strong relational integrity.

  Redis
  → Real-time caching for rewards.

- **ML Frameworks**
  PyTorch / TensorFlow
  → Custom model training.

  XGBoost
  → Efficient tabular prediction.

  Feast
  → Centralized feature store.

- **Infrastructure**
  - AWS / GCP
  - Kubernetes
  - Kafka
  - Docker

  → Scalability and container orchestration.

7. Proof of Concept (PoC)

7.1 PoC Description

The PoC implements:

- Receipt image upload
- OCR-based extraction
- Structured transaction storage
- Basic reward calculation
- Dashboard display of reward balance

Purpose:
Validate ML feasibility and reward processing workflow.

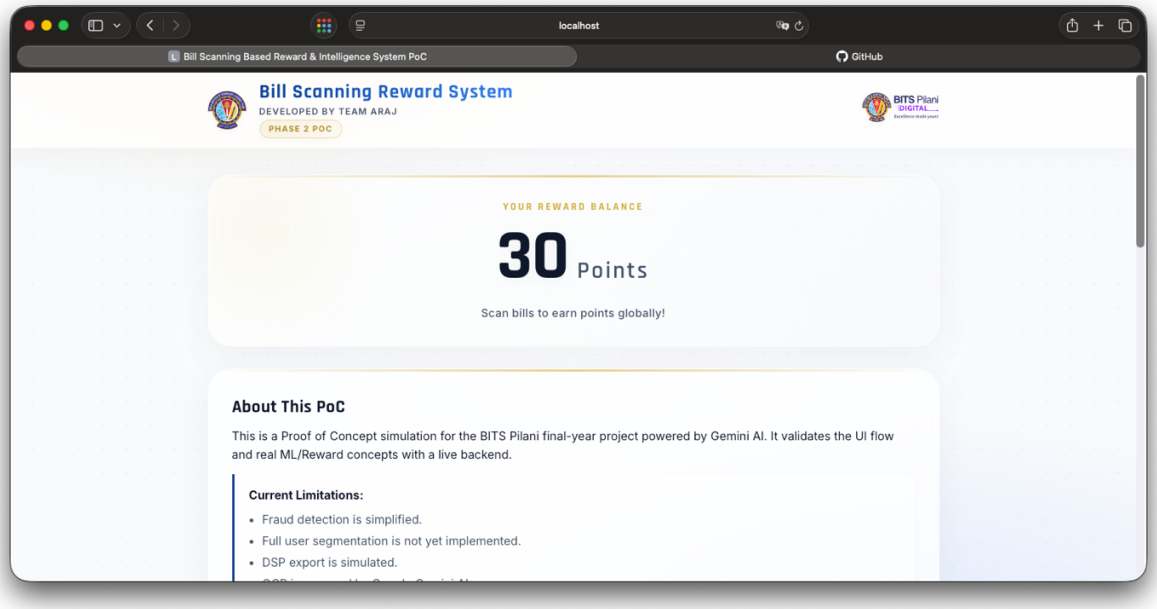
7.2 PoC Demonstration Details

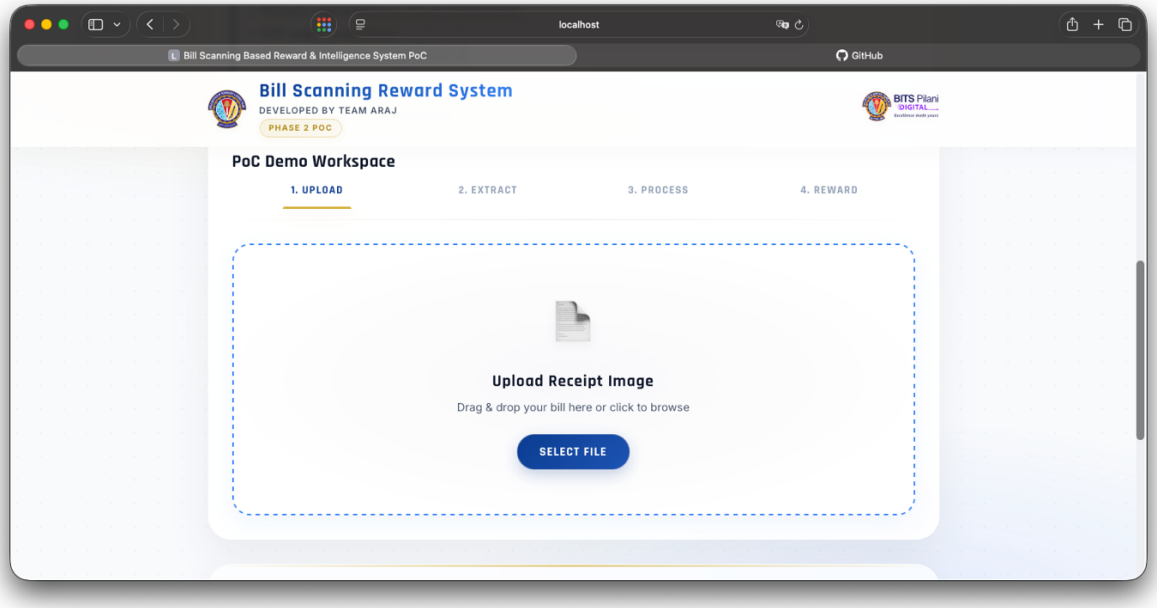
Demonstrated:

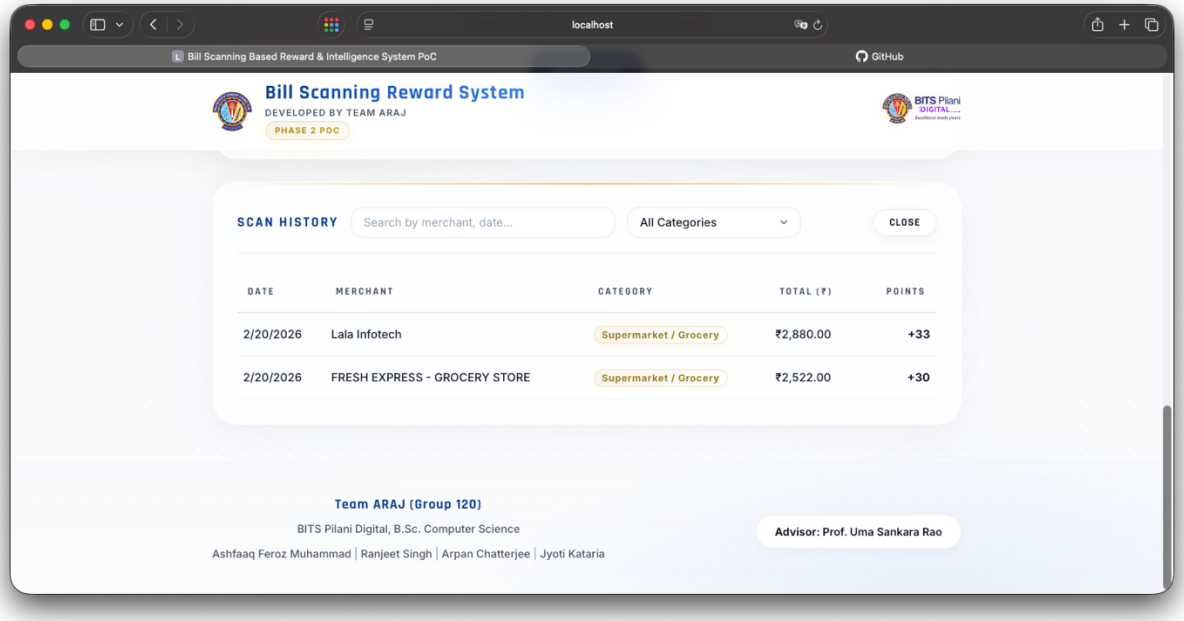
- Upload → Extract → Reward pipeline
- Merchant normalization
- Reward tier calculation

Limitations:

- Fraud detection simplified
- No full segmentation yet
- DSP export simulated


localhost
Bill Scanning Based Reward & Intelligence System PoC
GitHub
Bill Scanning Reward System
DEVELOPED BY TEAM ARAJ
PHASE 2 POC
BITS Pilani
YOUR REWARD BALANCE
30 Points
Scan bills to earn points globally!
About This PoC
This is a Proof of Concept simulation for the BITS Pilani final-year project powered by Gemini AI. It validates the UI flow and real ML/Reward concepts with a live backend.
Current Limitations:
Fraud detection is simplified.
Full user segmentation is not yet implemented.
DSP export is simulated.


localhost
Bill Scanning Based Reward & Intelligence System PoC
GitHub
Bill Scanning Reward System
DEVELOPED BY TEAM ARAJ
PHASE 2 POC
BITS Pilani
PoC Demo Workspace
1. UPLOAD
2. EXTRACT
3. PROCESS
4. REWARD
Upload Receipt Image
Drag & drop your bill here or click to browse
SELECT FILE



GitHub Respo Link: https://github.com/ashfaaqkt/Bill-Scanning-Based-Reward-Intelligence-System-study-project-bits-pop

8. Testing and Validation Strategy

Unit Testing:

- Service-level testing
- Reward calculation validation

Integration Testing:

- Receipt → ML → DB → Reward pipeline

Load Testing:

- Simulated concurrent uploads

ML Validation:

- OCR accuracy metrics
- Precision/Recall for classification
- Fraud model AUC score

9. Risks, Challenges, and Mitigation

Risk: OCR inaccuracy
Mitigation: Continuous model retraining

Risk: Fraud abuse
Mitigation: Real-time anomaly detection

Risk: Privacy concerns
Mitigation: Explicit consent logging + hashed exports

Risk: DSP integration complexity
Mitigation: Standardized cohort API formats

10. Phase 2 Outcomes and Readiness for Phase 3

Completed:

- System architecture finalized
- ML pipeline designed
- Core backend structure defined
- PoC implemented for ingestion and reward calculation

System is ready for:

- Full ML deployment
- Behavioral optimization
- DSP integration scaling
- Production hardening

The End
Thank You..

Advisor Feedback:
[To be filled]

Supervisor Comments:
[To be filled]

Recommendations:
[To be filled]

Submission Date	Group Advisor	TEAM
21-Feb-2026	Prof. Uma Sankara Rao	ARAJ